



Project page

witness lab, results, BibTeX
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LatentGauge: Planning-Geometry Attacks in a Measure-Preserving Blind Spot

New York University · Cornell University · Massachusetts Institute of Technology · Brown University

Same prediction · same latent law · **different ruler**

Interactive story

LeWorldModel → LatentGauge
[crcr0.github.io/
LatentGauge](https://crcr0.github.io/LatentGauge)



Motivation

Latent world models plan by comparing imagined futures with an encoded goal. Their learned representation is therefore not only a predictive state; it is also the coordinate system the planner uses to rank actions. Training constrains **prediction** and the **latent distribution**. Planning additionally trusts the **metric** of the coordinates handed to the scorer.

① prediction

Exact prediction retained

A conjugated pair $\tilde{h} = T \cdot h$, $\tilde{F} = T \cdot F \cdot T^{-1}$ keeps zero prediction risk; fixed-action rollouts deliver the same endpoints.

✓ check passes

② latent law

Gaussian marginal preserved

$T_{\#}\gamma = \gamma$ exactly. The controlled Gaussian diagnostic reads 0.40853 clean and 0.40910 mapped, within resampling variability.

✓ check passes

③ movement

Displacement within budget

$E\|T(Z) - Z\|^2 \leq \varepsilon^2$, yet at matched displacement local distortion $\|G_T - I\|_F$ ranges from 0.41 to 5.6 $\times 10^4$.

✓ check passes

— LatentGauge

The planner's metric changes

$G_T = DT^T DT$ enters the action-side Gauss–Newton matrix; endpoint–goal distances, rankings and CEM elite sets move.

what the planner actually uses

Core question. Which shared coordinate changes alter planning selection and updates while preserving the latent target law under a displacement budget?

Preliminary

Coordinate freedom. With $z_t = h(o_t)$ and $\hat{z}_{t+1} = F(z_t, a_t)$, a representable conjugacy

$$\tilde{h} = T \cdot h, \quad \tilde{F}(y, a) = T(F(T^{-1}y, a))$$

keeps exact prediction and, when $h(o_t) - \gamma = N(o, I_d)$ and $T_{\#}\gamma = \gamma$, the marginal too, while distances can change.

Planning geometry. For the endpoint $\Phi(a)$, goal z_g and $J = D_a \Phi$:

$$C(a) = \|\Phi(a) - z_g\|^2, \quad H_o^{GN} = 2J^T J$$

$$C_T(a) = \|T(\Phi(a)) - T(z_g)\|^2, \quad H_T^{GN} = 2J^T G_T(\Phi(a))J$$

Proposition (CEM boundary selection)

For N fixed candidates and elite sets E, E_T of size k , with clean rank in parentheses: $C_T(a_{(k+1)}) < C_T(a_{(k)}) \Rightarrow E_T \neq E$. Reordering inside the elite set leaves its moments unchanged.

Threat model

The attacker reads calibration endpoints and goals and fits **one shared differentiable bijection after prediction**. Encoder, predictor, observations, goal input and scalar scorer stay fixed; the map is reused across goals, candidates and calls. Fixed-action rollouts are unchanged; later CEM proposals and MPC actions may change.

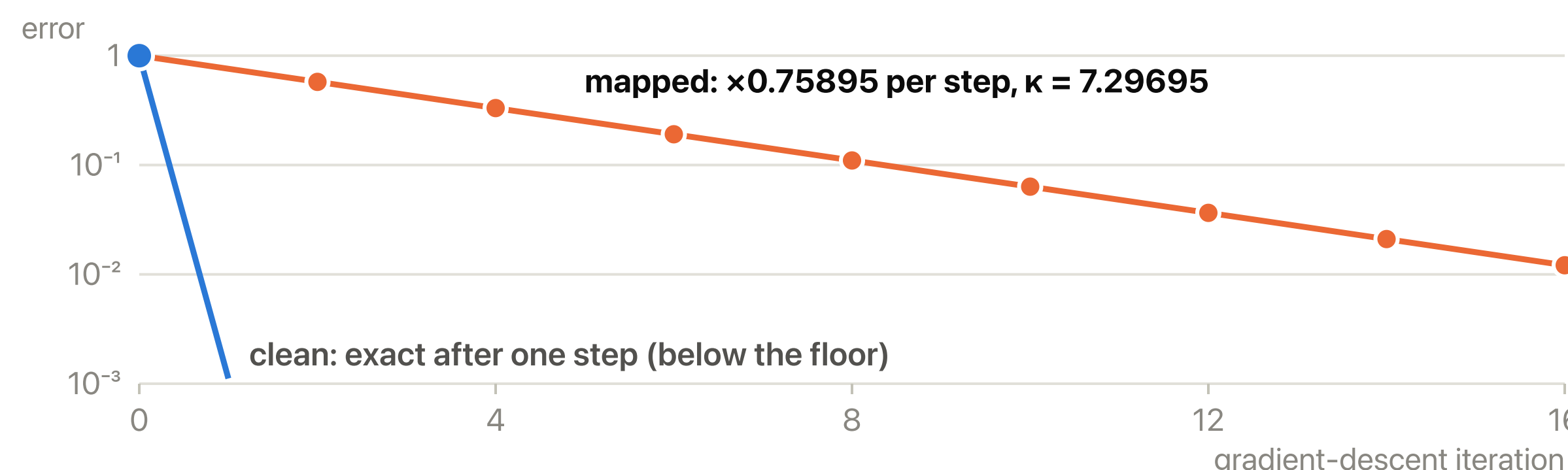


Figure 2. One Gauss–Newton channel: with J fixed, the map raises the active conditioning from 1 to 7.29695, so exact gradient descent contracts at $(\kappa - 1)/(\kappa + 1)$ per step.

Contributions

1. We formulate inference-time planning interventions inside a **measure-preserving coordinate space**.
2. We connect predictive coordinate freedom, local action geometry and exact candidate costs to a **margin-resolved bound on successive CEM updates**.
3. We construct a **shared scoring map** whose native-boundary calibration feeds kernel extension and budgeted profile fitting.
4. We validate the construction through **exact ranking, displacement and invariance calculations**, and specify paired planning traces for task-level evaluation.

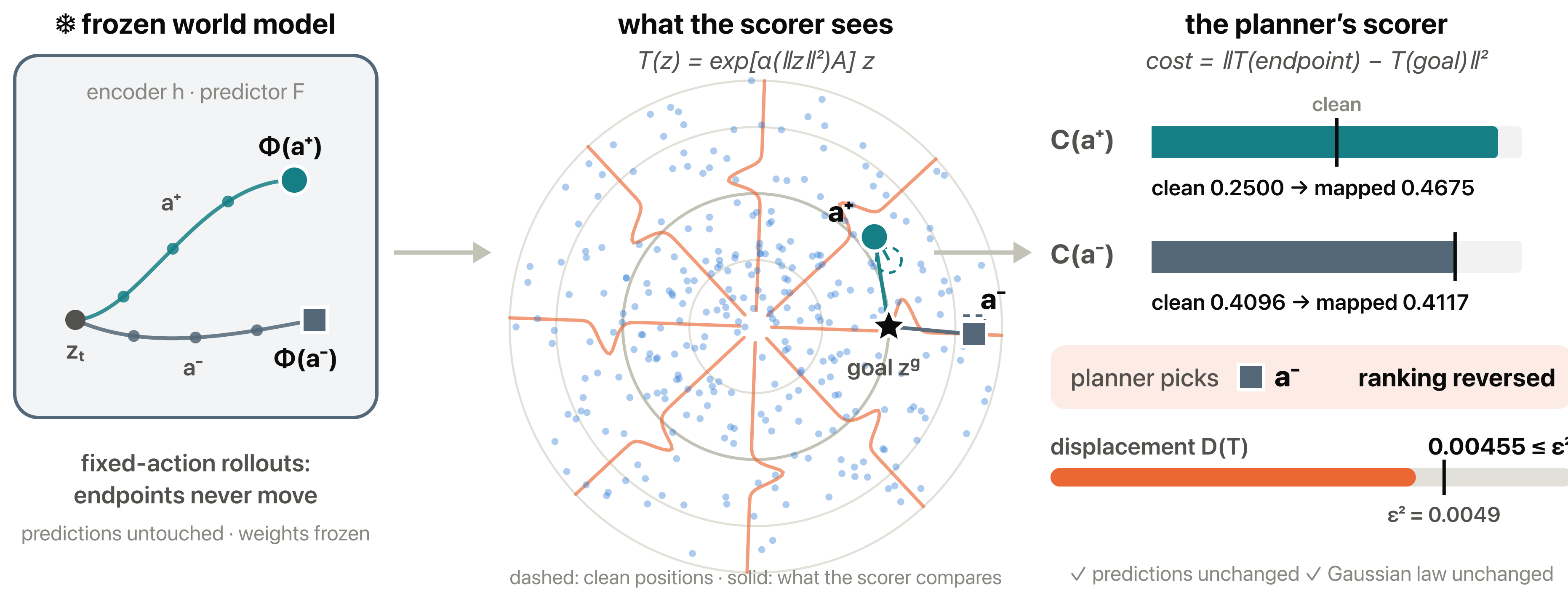


Figure 1. Same prediction, same marginal, different ruler. The frozen world model's endpoints never change. LatentGauge rotates each Gaussian shell by its own angle $\alpha(\|z\|^2)$: radii and the latent law are preserved and the goal shell stays fixed ($\alpha(1) = 0$), yet the scorer's geometry shears and the planner's preference flips from a^+ to a^- with displacement $0.00455 < \varepsilon^2 = 0.0049$ (exact canonical witness).

Method

1

Conservation fixes the form

Gaussian conservation requires $\text{div } v - z^T v = 0$; changing geometry needs nonrigid motion. A radius-dependent rotation does both, in closed form:

$$T_c(z) = e^{\alpha_c(\|z\|^2)A} z, \quad A^T = -A$$

a radius- and Gaussian-preserving bijection with inverse $e^{-\alpha_c(\|y\|^2)A} y$.

what is admissible

2

The native CEM boundary calibrates

Adjacent candidates across the elite boundary (clean ranks $k, k+1$) form calibration triples; their squared radii become kernel sites:

$$\alpha_c(s) = \sum_j c_j K(s, s_j)$$

The leading block of the skew first variation selects the rank-two plane A .

where to push

3

Exact scores drive the fit

Minimize the mean softplus of mapped boundary margins under the budget, by feasible trust-region SQP from the identity:

$$\min_c (1/n) \sum_i \text{softplus}(m_i^T) \text{ s.t. } D_{\min}(c) \leq \varepsilon^2$$

The score-invariant common rotation is removed exactly: $b^* = -\arg M_c$. $D_{\min} = 4(1 - |M_c|)$.

how far to go

$$C_T(a) - C(a) = 2 z_H^T [I - e^{\Delta A}] z_g$$

exact cost shift, $\Delta A = \alpha_c(\|z_g\|^2) - \alpha_c(\|z_H\|^2)$: equal endpoint and goal radii give zero shift; a common angle changes no score

$$T_{\#}\gamma = \gamma, \quad D_{\gamma}(T) = E_{Z \sim \gamma} \|T(Z) - Z\|^2 \leq \varepsilon^2$$

feasible set: the reference law is preserved exactly and the average displacement is budgeted ($\varepsilon = 0.07$)

Algorithm 1 Shared Gaussian-preserving scoring, Score

1. **Require** $\{z_i\}_{i=1}^N, z_g, \alpha, A$
2. $\tilde{z}_g \leftarrow e^{\alpha(\|z_g\|^2)A} z_g$
3. **for** $i = 1, \dots, N$ **do**
4. $\tilde{z}_i \leftarrow e^{\alpha(\|z_i\|^2)A} z_i; C_i^T \leftarrow \|\tilde{z}_i - \tilde{z}_g\|^2$
5. **return** $(C_i^T)_{i=1}^N$

Algorithm 2 Planner-aligned kernel-profile fitting

1. **Require** calibration bank $\mathcal{B}$; budget ε
2. boundary triples $(z_i^+, z_i^-, z_{g,i}) \leftarrow \text{Pairs}(\mathcal{B})$; plane A , sites, kernel $\leftarrow \text{Calibrate}$
3. $c \leftarrow 0$; **while** call allowance remains **do**
4. score pairs with **Score**; exact loss and budget derivatives
5. trust-region SQP step, feasibility restoration, acceptance
6. $b \leftarrow -\arg E_{S \sim \chi^2_{d+2}} e^{i\alpha_c(S)}$; **return** $T(z) = e^{\alpha_c(\|z\|^2) + b\|A} z$

Prediction, marginal validity and the geometry used for decisions are **distinct interface properties**.

Results

Residual success · LeWM / PushT (CEM)

96% → **62.40%**

−15.36 pts vs equal-budget random-direction search

Residual success · DINO+TS / PushT (GD/MPC)

85.33% → **58.02%**

−12.80 pts vs random-direction search

Top-one FlipRate (orthogonal control: 0%)

0% → **56%**

50% on DINO+TS / PushT

Elite replacement · native CEM

9% → **22%**

boundary vs top-one calibration

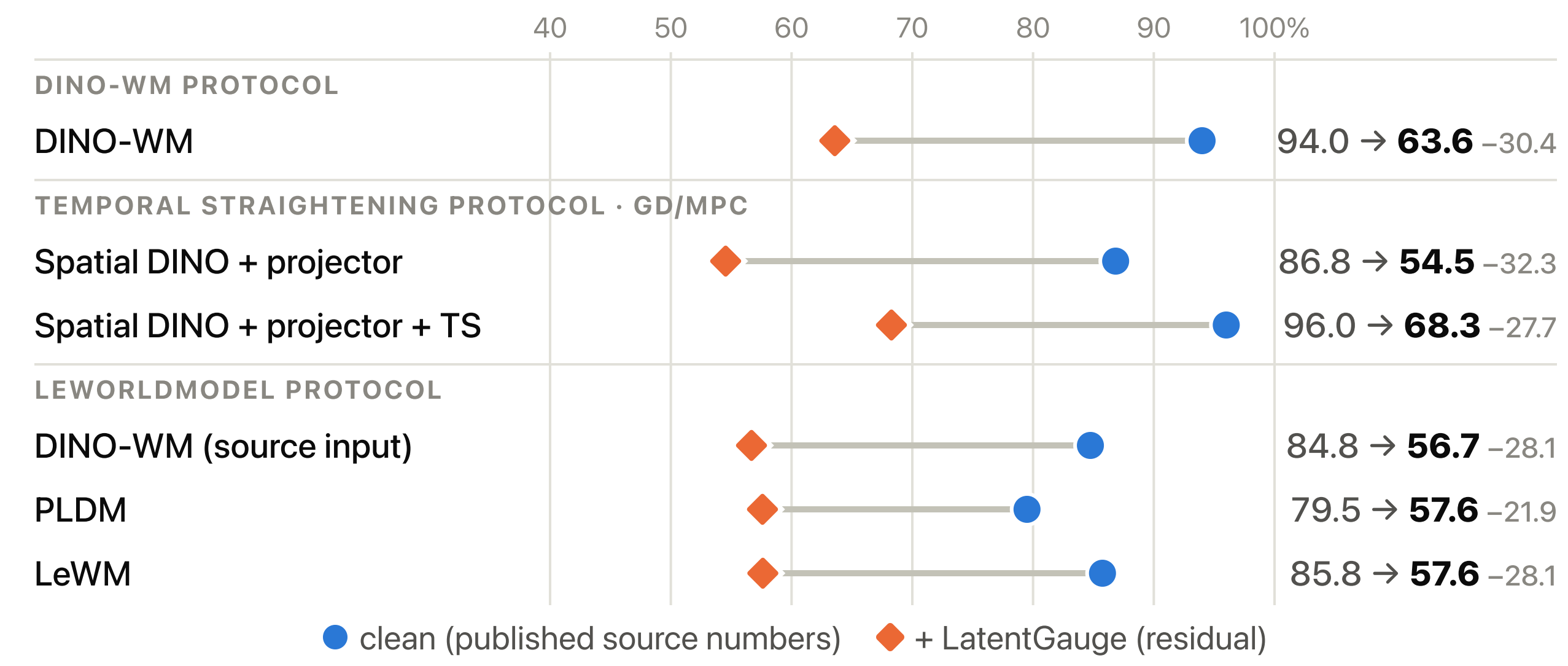


Figure 3. Mean success over each protocol's four tasks (%), $\varepsilon = 0.07$.

	LeWM / PushT (CEM)			DINO+TS / PushT (GD/MPC)		
Shared map (exact γ law)	SR ↓	Flip ↑	TPR ↓	SR ↓	Flip ↑	TPR ↓
Identity	96	0	1	85.33	0	1
Orthogonal Q	96	0	8	85.33	0	10
Random kernel profile	88.32	14	12	79.36	12	14
Random-direction search	77.76	32	13	70.82	29	15
LatentGauge (pairwise)	62.40	56	15	58.02	50	18

Table 1. Shared-map controls. SR: residual task success; Flip: fixed-bank top-one replay reversal; TPR: learned-feature two-sample detection at nominal 1% FPR (all %).

Exact witness & mechanism checks

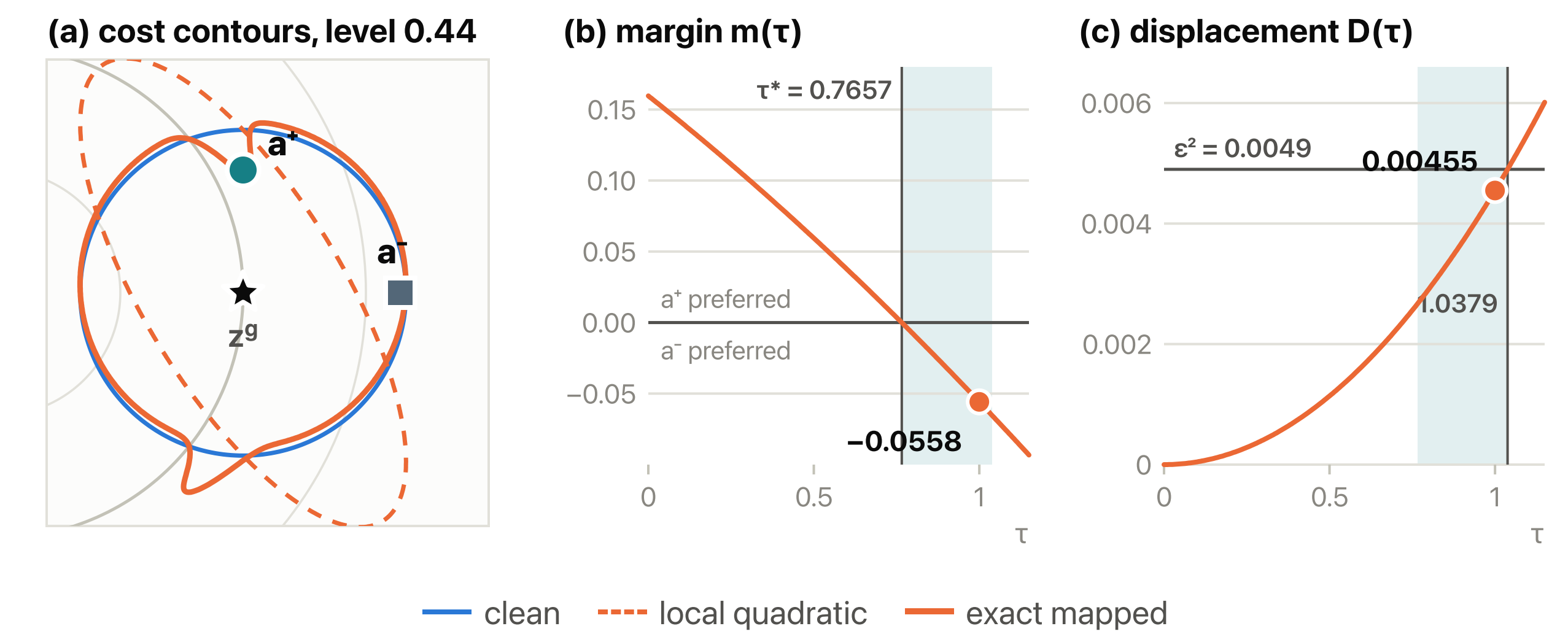


Figure 4. One exact witness: the margin crosses zero at $\tau^* = 0.765739$ and $D_{\gamma}(T_{\tau}) = 0.004548694 < 0.0049$ (shaded: strict reversal within budget).

- **Boundary, not ranking.** Coupled CEM runs chose different elite sets in **476 of 512** rounds under the radial map, none under identity or a common rotation; every propagation envelope held.
- **Invisible to the marginal audit.** 20 Gaussian batches: paired change 0.000575, 95% interval $[-0.000466, 0.001616]$. A common rotation, or normalizing to one shell *before* the map, preserves every ordering.

References

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